Practice Problem 1

Consider a tube filled with nitrogen gas but closed on one end. If at the temperature of 22°C its fundamental frequency is 200 Hz, determine

a) the length of the tube.

Solution:

$T = 22°C$

Frequency of the nth harmonic (normal mode f)

Nitrogen, N_2

$n = 1 \rightarrow f = 200\,Hz$

$$f_n = \frac{nV}{4L} \quad \text{at } n = 1, \; f_1 = 200\,Hz$$

Calculate speed of sound at 22°C by assuming N_2 is an ideal gas: $V = \sqrt{\frac{\gamma RT}{M}}$.

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Solution (Con't)

$$V = \sqrt{\frac{\gamma R T}{M}}$$

$R = 8.31 \text{ J/mol-K}$

γ (diatomic) $= 1.4$ ← Heat Capacity Ratio

$M \rightarrow$ molar mass (kg/mol)

$$M_{N_2} = 2M_N = 2(14 \text{ g/mol})$$

$$= 0.028 \text{ kg/mol}$$

$$\therefore V = \sqrt{\frac{(1.4)(8.31)(22+273)}{0.028}} \text{ m/s}$$

$$\approx 350.1 \text{ m/s}$$

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Solution (con't)

$$f_n = \frac{nv}{4L} \qquad n=1 \rightarrow f_1 = 200\,Hz$$

$$L = \frac{v}{4f_1} = \frac{(350.1)}{4(200)}\ m$$

$$L = 0.4376\ m$$

b) Determine the frequency of the 3rd harmonic

Solution.

At the 3rd harmonic $\rightarrow n = 5$

$$\therefore f_5 = \frac{5v}{4L}$$

speed of sound, $v = 350.1\ m/s$

in N_2

$$L = 0.4376\ m$$

$$f_5 = \frac{5(350)}{4(0.4376)}\ Hz$$

$$\approx \underline{1000\ Hz}$$

Practice Problem 2

Sulphur dioxide is an intermediate in the industrial production of sulphuric acid.

Consider a closed tube filled with gaseous sulphur dioxide.

If at the temperature of 30°C its fundamental frequency is 100 Hz, determine

a) the length of the tube.

b) the frequency of its 2nd harmonic.

Solution.

a) $f_n = \frac{nV}{2L}$; $n=1 \rightarrow f_1 = 100\ Hz$

Speed of sound in SO_2.

$$V = \sqrt{\frac{\gamma RT}{M}}$$

$R = 8.31\ J/mol\text{-}K$

γ (triatomic) $= 1.3$

↑ Heat Capacity Ratio

$M \rightarrow$ molar mass (kg/mol)

$\gamma = 1.3$

$M = m_s + 2(m_o) = [32 + 2(16)]\ g/mol$

$= 0.064\ kg/mol$

$$V = \sqrt{\frac{(1.3)(8.31)(30+273)}{0.064}}\ m/s$$

$\approx 226.15\ m/s$

Solution [Con't]

$$f_1 = \frac{V}{2L} \rightarrow L = \frac{V}{2f_1}$$

$$L = \frac{226.15}{2(100)}\ m = \underline{\quad 1.13\ m \quad}$$

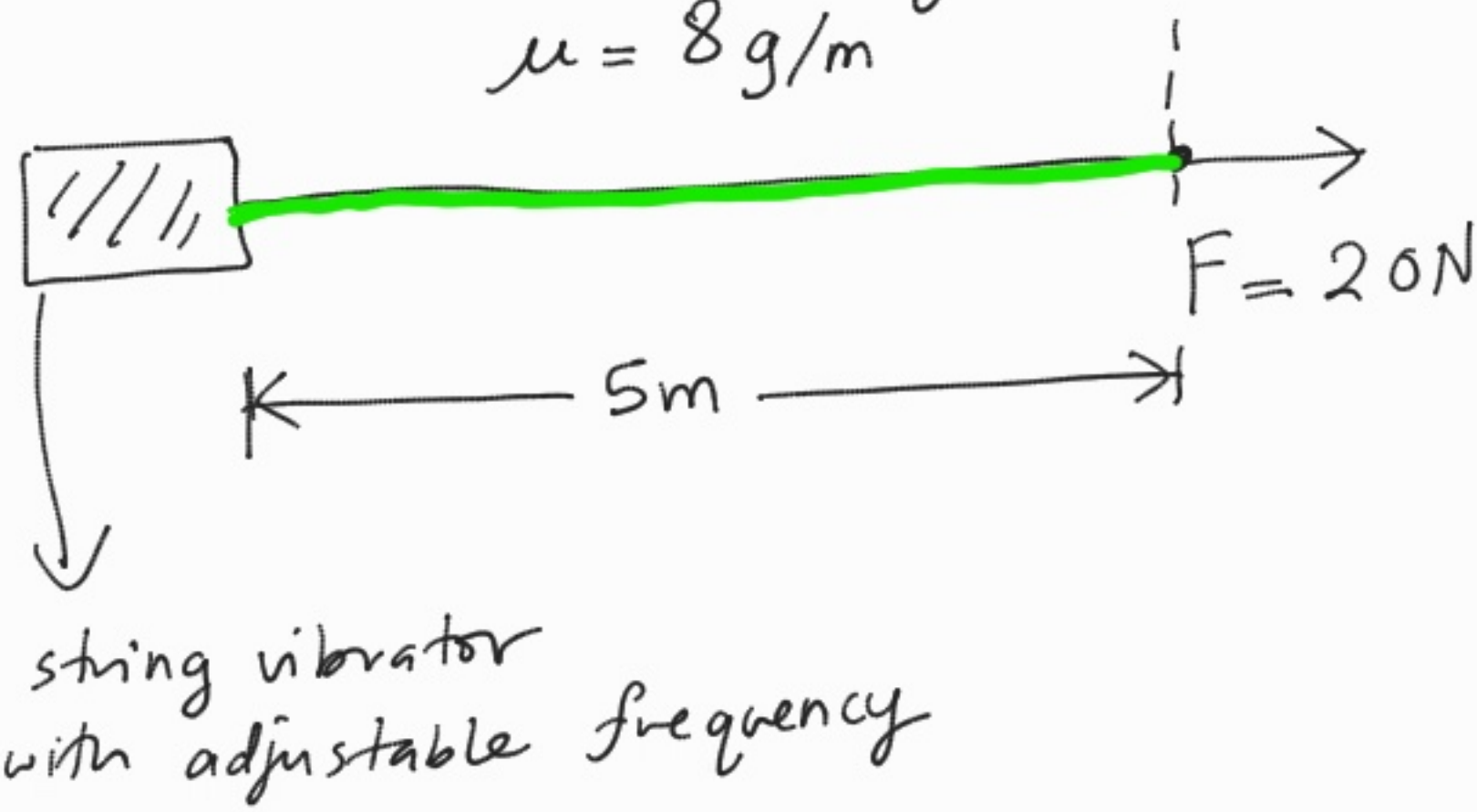
b) $n = 2 \rightarrow f_2 = ?$ [Both ends closed]

$$f_2 = \frac{V}{L} = \frac{226.15}{1.13}\ Hz$$

$$f_2 \approx \underline{\quad 200\ Hz \quad}$$

Practice Problem 3

Using the following setup determine the velocity of the waves on the string.



Note: waves generated by the string vibrator travel on the string & are reflected by the fixed boundary condition at the end of the string.

Solution.

$$\text{Tension, } T = F = 20\,N$$

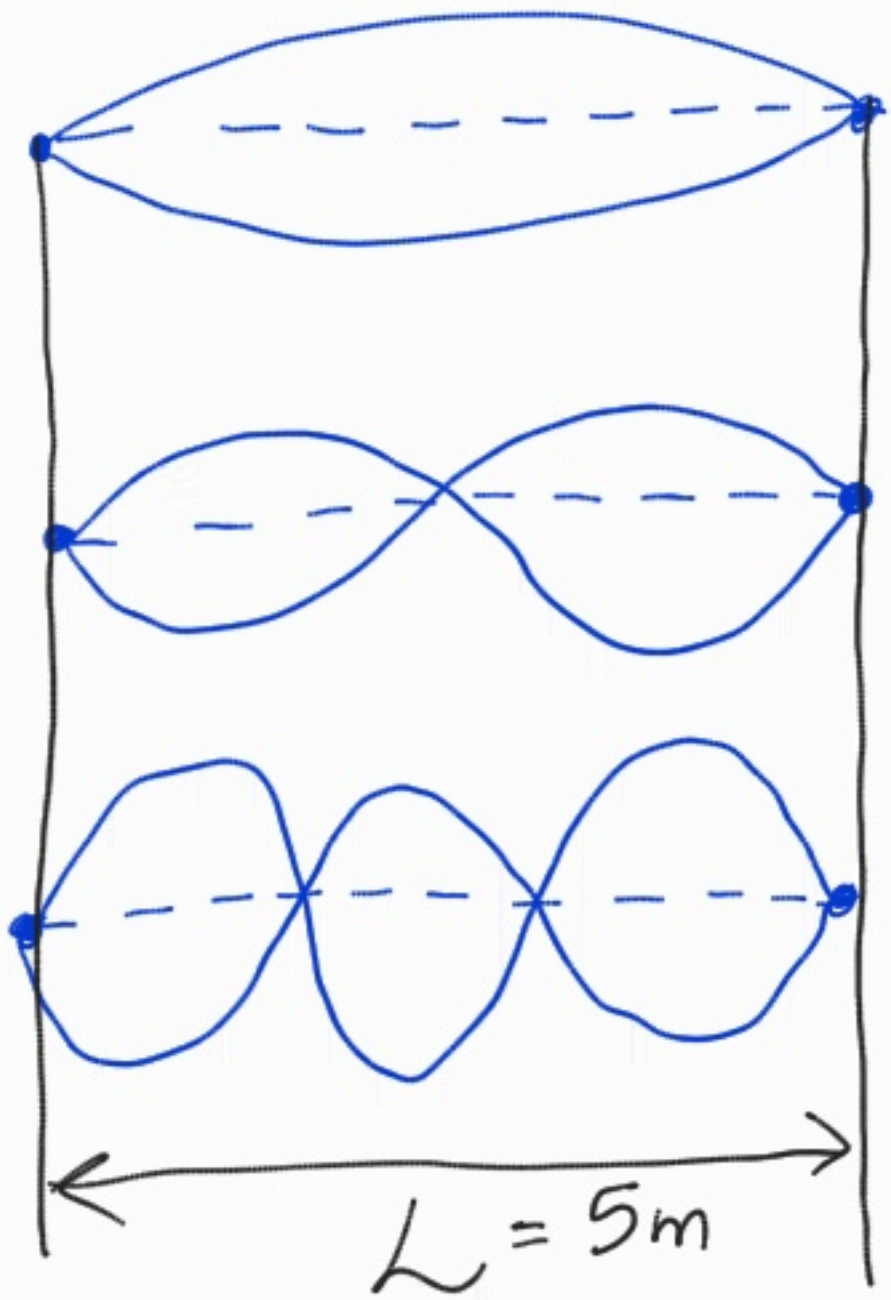
$$\text{velocity, } V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{20\,N}{0.008\,kg/m}}$$

$$= 50 \quad m/s$$

b) Sketch the standing waves for the 1st 3 modes generated & provide the expressions for the wavelength with respect to the string length.

c) Calculate the frequencies that need to be set in the string vibrator so it produces the 1st 3 modes of the standing waves.

Solution.



$$f_n = \frac{nV}{2L}$$

$$f_1 = \frac{50}{2(5)} = 5\ Hz$$

$$f_2 = \frac{2(50)}{2(5)} = 10\ Hz$$

$$f_3 = \frac{3(50)}{2(5)} = 15\ Hz$$